Object-Oriented Software Engineering & SOLID Principles

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Core Object-Oriented Pillars

? Encapsulation: Restricting direct access to object state and coupling data with mutating operations.
? Abstraction: Presenting simplified, high-level interfaces while concealing implementation complexities.
? Inheritance: Mechanism where child classes derive properties and behavior from generalized parent classes.
? Polymorphism: Ability of different objects to respond to the same message with customized behavior.

2. The SOLID Principles for Maintainable Code

? S ? Single Responsibility Principle (SRP): A class should have one, and only one, reason to change.
? O ? Open/Closed Principle (OCP): Software entities should be open for extension, but closed for modification.
? L ? Liskov Substitution Principle (LSP): Subtypes must be substitutable for their base types without altering system correctness.
? I ? Interface Segregation Principle (ISP): Clients should not be forced to depend on interfaces they do not use.
? D ? Dependency Inversion Principle (DIP): Depend upon abstractions, not concretions. High-level modules should not depend upon low-level modules.

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